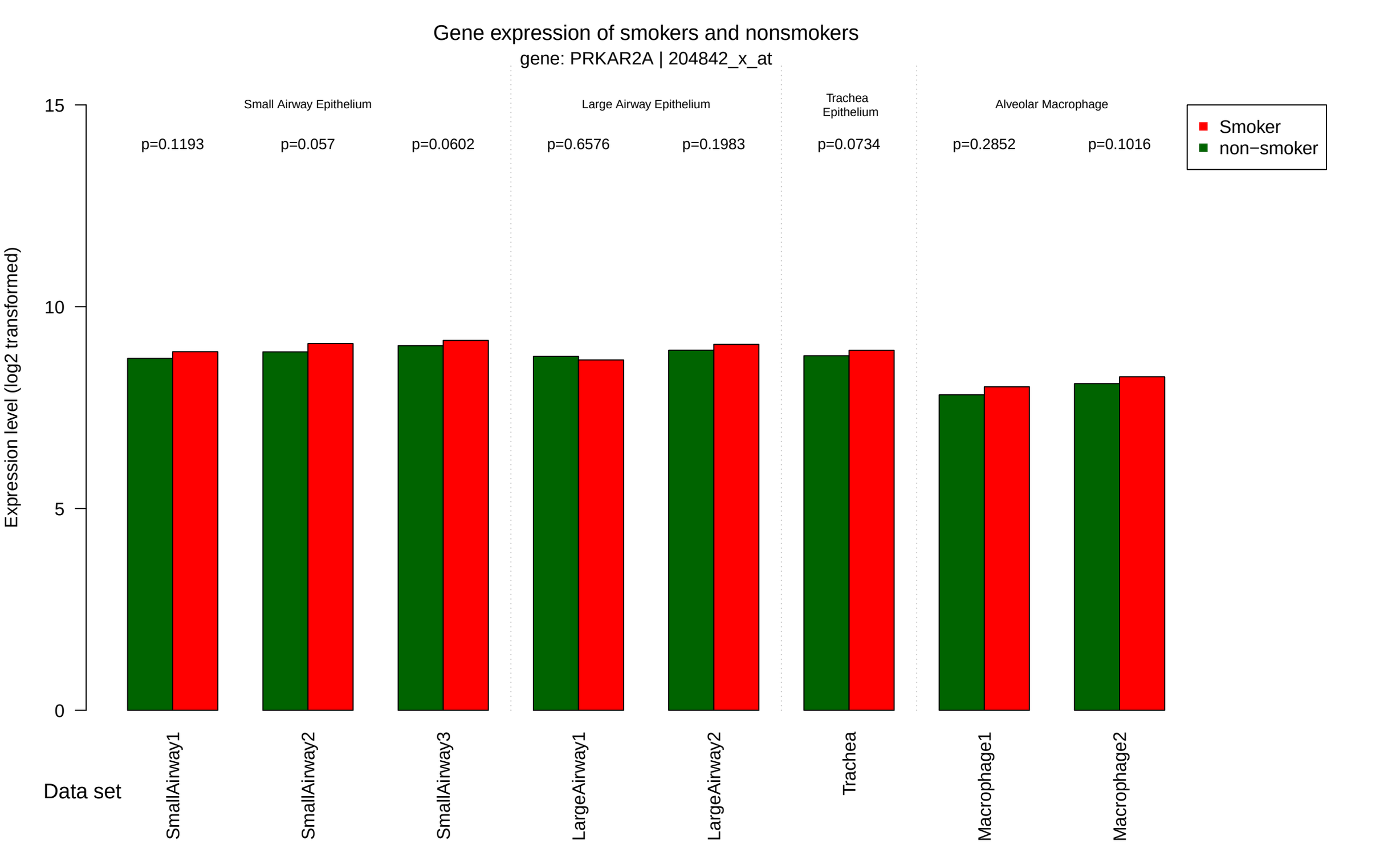
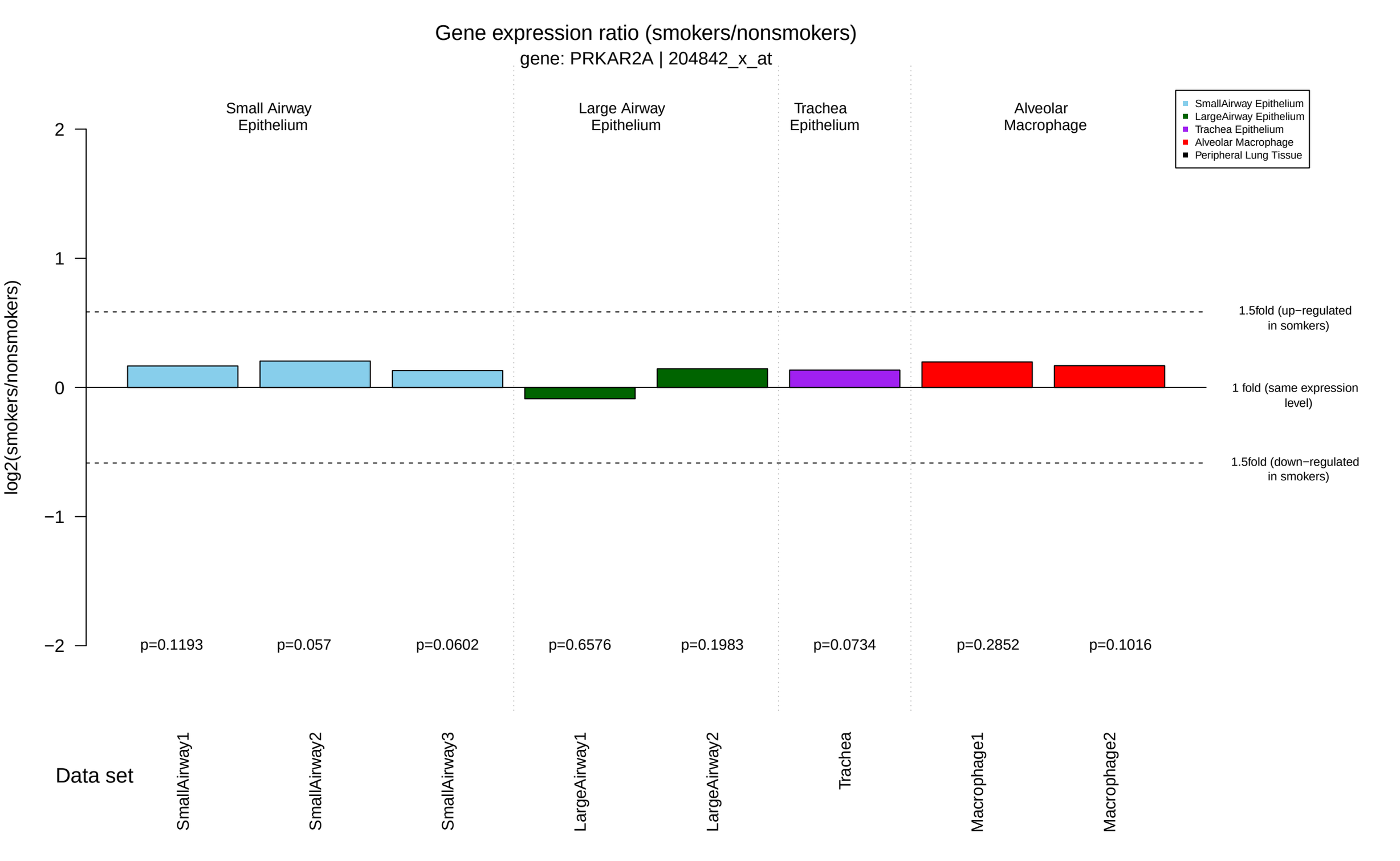
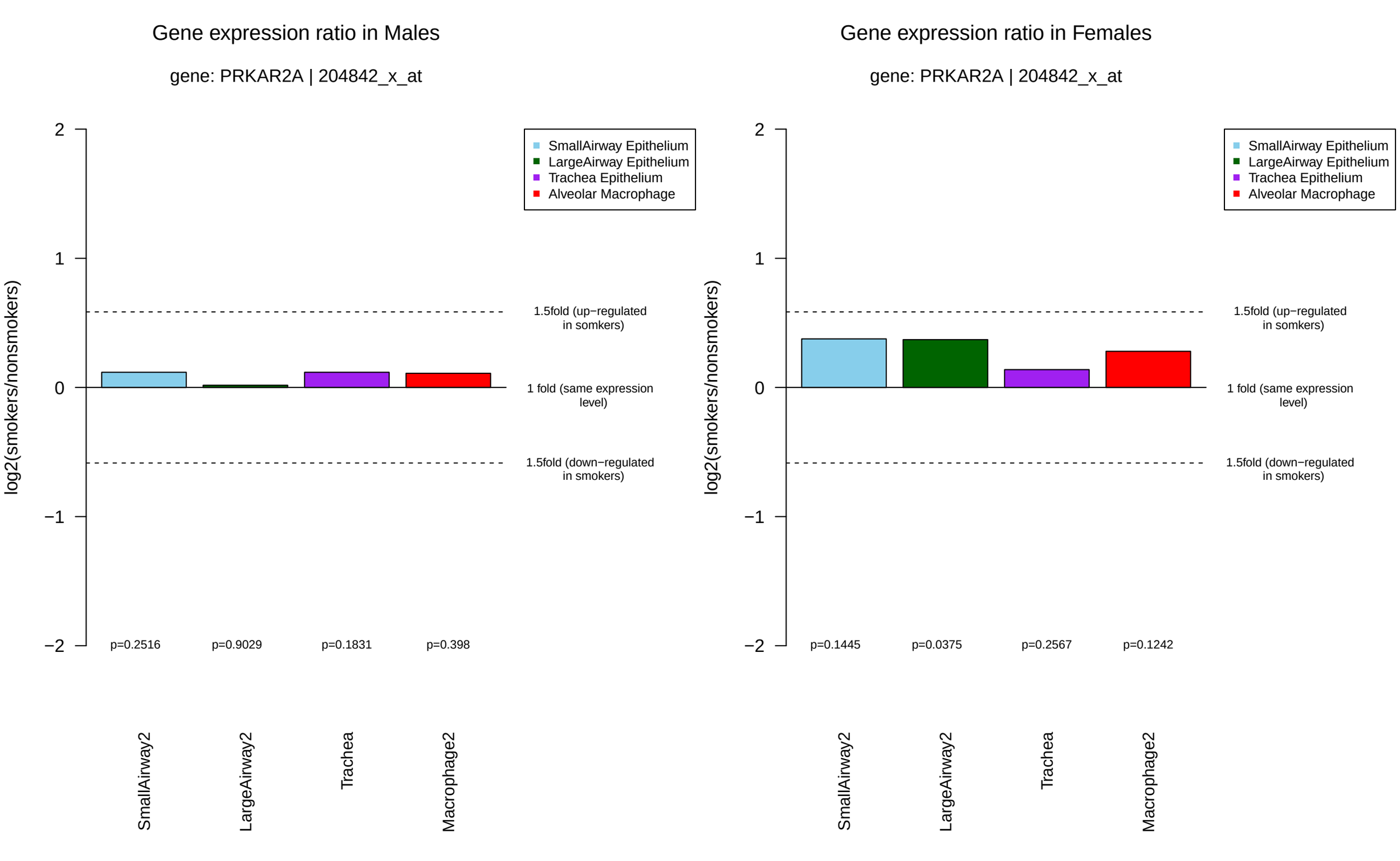
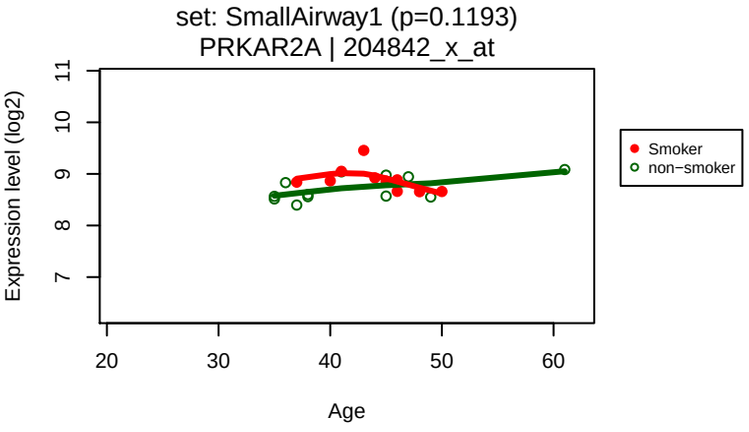
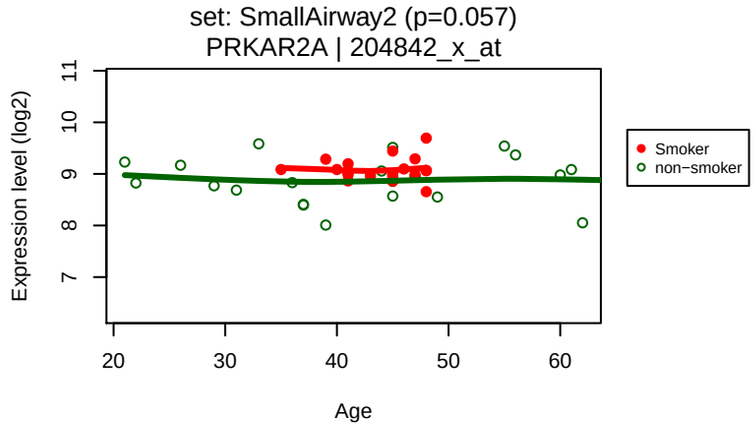
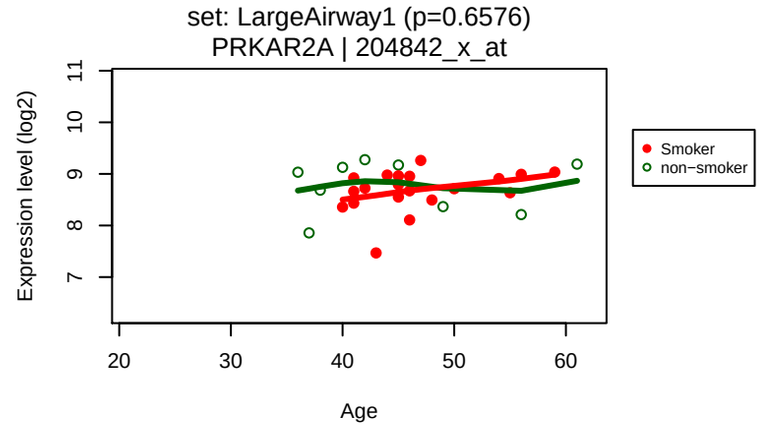

Gene expression of smokers and nonsmokers
gene: PRKAR2A | 204842_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1193
p=0.057
p=0.0602
p=0.6576
p=0.1983
p=0.0734
p=0.2852
p=0.1016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

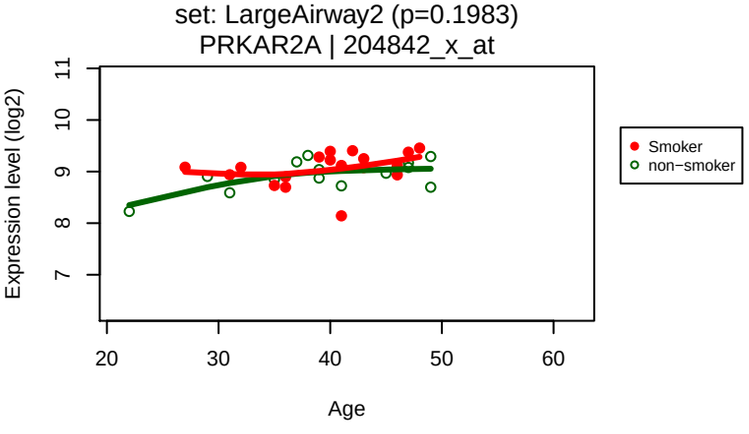

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR2A | 204842_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1193
p=0.057
p=0.0602
p=0.6576
p=0.1983
p=0.0734
p=0.2852
p=0.1016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

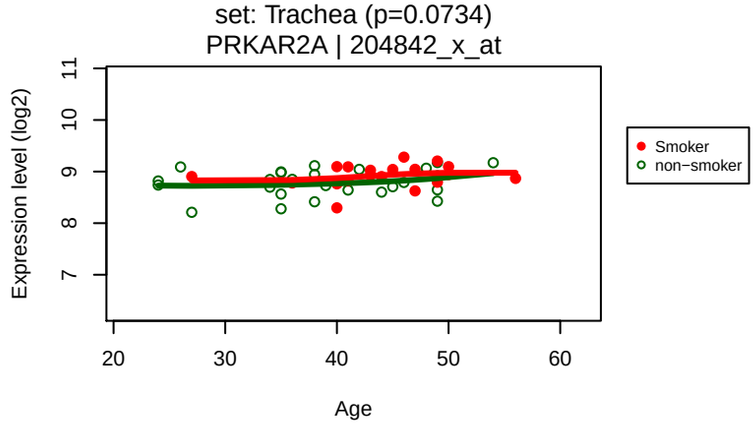

Gene expression ratio in Males
gene: PRKAR2A | 204842_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2516
p=0.9029
p=0.1831
p=0.398
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR2A | 204842_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1445
p=0.0375
p=0.2567
p=0.1242
SmallAirway2
LargeAirway2
Trachea
Macrophage2

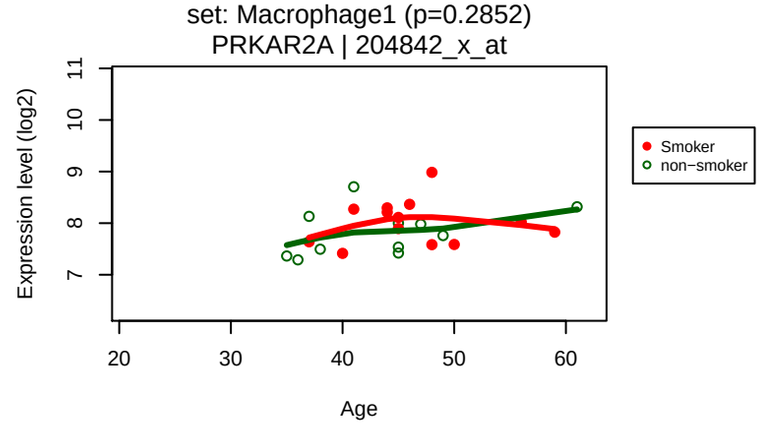

set: SmallAirway1 (p=0.1193)
PRKAR2A | 204842_x_at
Expression level (log2)
Age
Smoker
non−smoker

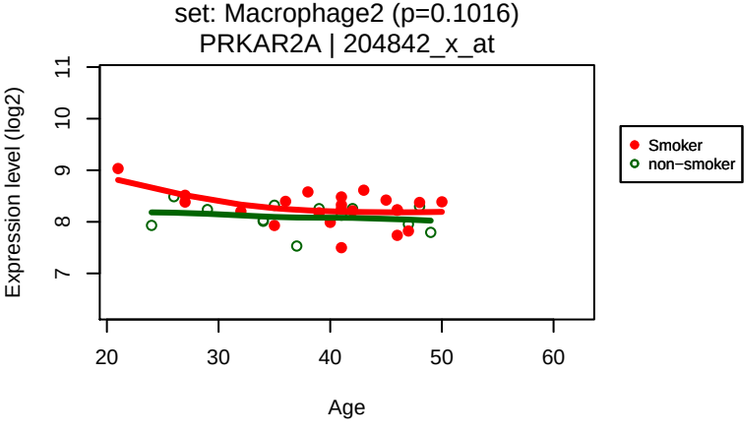

set: SmallAirway2 (p=0.057)
PRKAR2A | 204842_x_at
Expression level (log2)
Age
Smoker
non−smoker

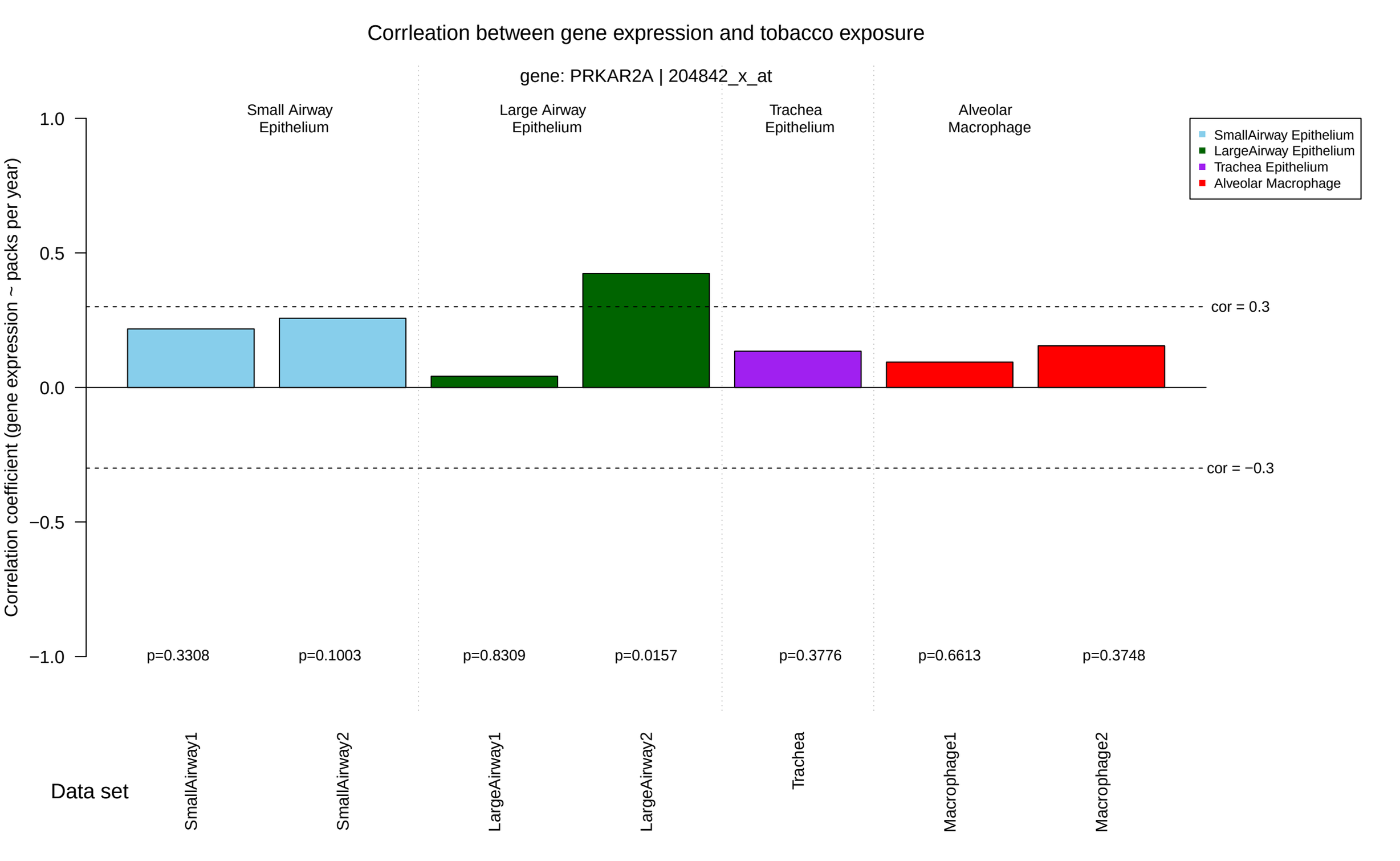

set: LargeAirway1 (p=0.6576)
PRKAR2A | 204842_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1983)
PRKAR2A | 204842_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0734)
PRKAR2A | 204842_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2852)
PRKAR2A | 204842_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1016)
PRKAR2A | 204842_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR2A | 204842_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3308
p=0.1003
p=0.8309
p=0.0157
p=0.3776
p=0.6613
p=0.3748
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2